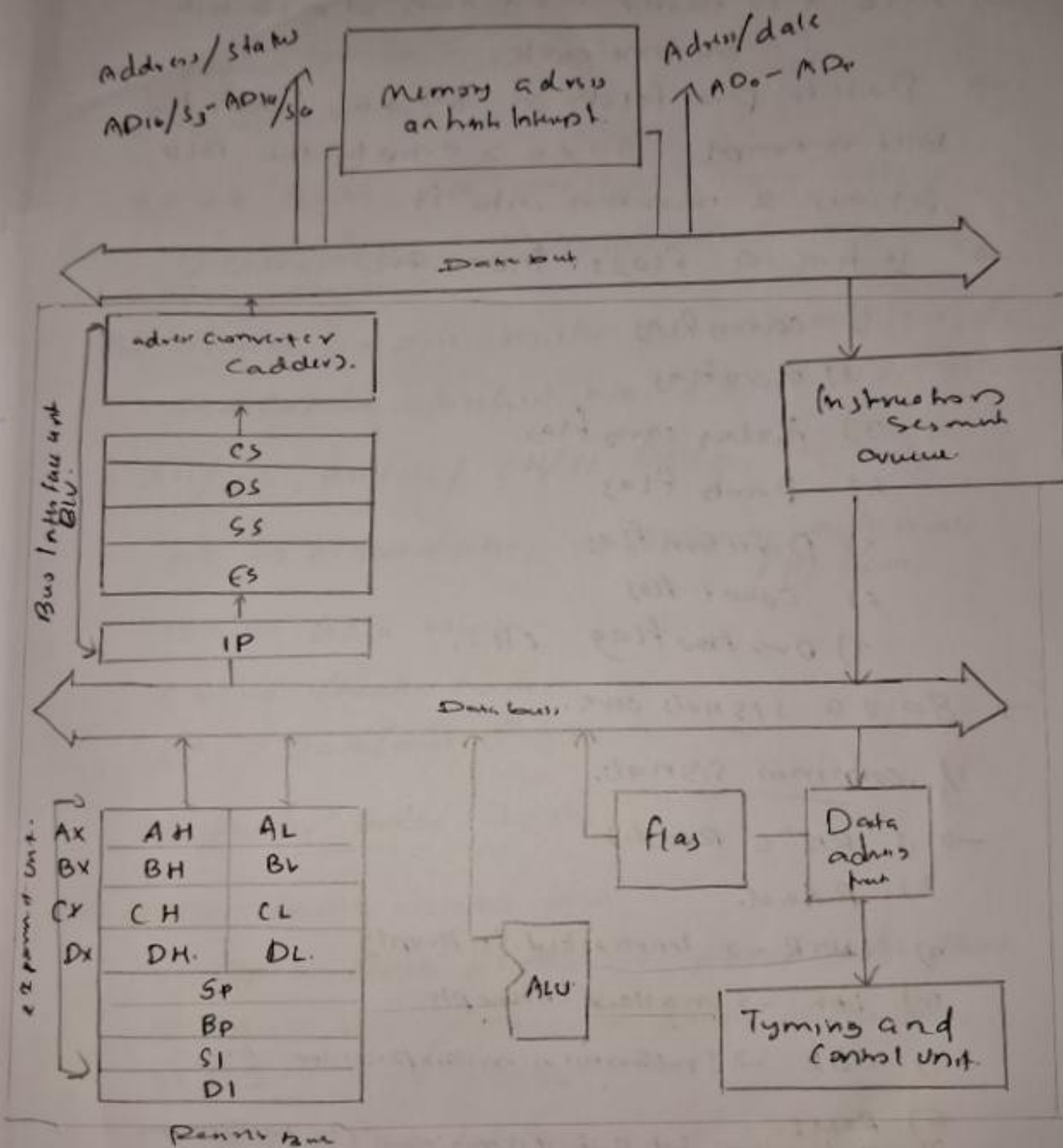


6) a).

8086 Architecture



- 8086 is a 16 bit microprocessor.
- It has 16 bit address bus.
- It is a pipelined architecture.
- 8086 has memory banking since it is

16 bit data bus.

→ Also it is faster as it can do 16 bit in one cycle.

→ There is pre fetch in 8086, when the BIU is empty 8086 is empty the BIU fetches 2 numbers into it.

→ It has 9 flags. Those are.

1) carry flag

2) zero flag

3) Auxiliary carry flag

4) parity flag

5) Direction flag

6) Count flag

7) overflow flag etc...

→ 8086 signals are.

1) Common signals.

→ 1) RD['] (Reads)

2) Read.

3) INT_R → Unmasked Interrupts

4) INT_N → masked Interrupts.

5) CLK → Synchronizes microprocessor time

6) Reset.

7) min/max: → Set 0 if it is minimum else maximum

2) Special minimum signals

→ 1) INT_A

2) Lock

3) Hold, HLD_A

3) Special maximum signals

1) status lines $\rightarrow S_0, S_1, S_2$.

2) Data $\rightarrow D_1, D_2$ etc....

$\rightarrow$ 8086 have 16 registers. These are-

1) General purpose Register

The general purpose registers are AX, BX, CX, DX and can be classified into 8 bit registers as,

$\rightarrow$ AH/AL, BH/BL, CH/CL, DH/DL.

AX $\rightarrow$ Accumulator. $\rightarrow$ performs arithmetic operation,

BX $\rightarrow$ Base register.

CX $\rightarrow$ Counter register

DX $\rightarrow$ Data/Data Register

2) Pointer Index Register

The pointer index registers are

1) ESP $\rightarrow$ Stack pointer $\rightarrow$ points to the stack.

2) BP $\rightarrow$ Base pointer $\rightarrow$ points to the base

3) SI $\rightarrow$ Source Index

4) DI $\rightarrow$ Data Index.

3) Segment registers

MC. Segment registers are

1) CS $\rightarrow$ Code segment

2) DS $\rightarrow$ Data segment

3) SS $\rightarrow$ Stack segment

4) Extra arguments $\rightarrow$ points to the extra value

$\rightarrow$ The instruction set in 8086 are.

1) Data transfer instructions

1) $\text{MOV Dest, Src} \rightarrow$ Move (copy) contents of destination to source code.

eg: MOV AX, BX .

2) $\text{PUSH Src} \rightarrow$ Push into source into stack.

eg: PUSH AX .

3) $\text{POP Src} \rightarrow$ POP source ~~into~~ from stack into destination.

eg: POP AX .

2) Arithmetic instructions

1) $\text{ADD Dest, Src} \rightarrow$ Addition of 2 values

eg: ADD AX, BX

2) $\text{MUL Ax} \rightarrow$ Multiplication of value

eg: MUL AX .

3) $\text{DIV Ax} \rightarrow$ Division of value

eg: DIV AX

4) $\text{IN AL/AX Port} \rightarrow$ Port from destination to port.

AL/AX 80H.

67b).

8086

- 16 bit microprocessor
- 16 bit ^{data} ~~address~~ bus
- 9 flags
- It has memory banking
Since it has 16 bit ^{data} ~~address~~ bus.
- Faster since it can complete 16 bit in one cycle
- Supports pipelined architecture.
- Pipeline architecture is 6 bit.
- Can retch data, the BIU allows 2 number when it is empty.
- ~~Address lines~~ It has I/O to connect between memory and data.

8088.

- 16 bit microprocessor.
- 8 bit ^{data} ~~address~~ bus.
- 9 flags.
- It does not require memory banking since it has only 8 bit.
- Slower since it can complete only 8 bit in one cycle.
- Supports pipelined architecture.
- Pipeline architecture is 2 bit.

Can retch data, the BIU allows one number when it is empty.

- It has I/O to connect b/w memory and data.

8)

CLC

MOV SI 1000

MOV DI 2000

~~MOV~~
LEA AX

MOV AX (SI)

INC SI

INC SI

79) a) CMP → compare instruction

→ CMP compare two numbers/operands and applies the condition.

eg: CMP AX, BX.

b) MUL → multiplication instruction

→ MUL is the multiplication instruction in Arithmetic Instruction Set.

→ It is an instruction to multiply two numbers.

eg: MUL AX.

c) DIV → Division instruction

→ DIV is the division operation that is in Arithmetic Instruction Set.

→ It is an instruction to ~~multiply~~ ^{for} two divisions of numbers.

d) RCL → Right counter shift.

→ It rotates arithmetic right shift.

e) ROR → Right Rotate Shift.
ex: ROR 20

→ It also rotates to arithmetic left and right shift.

ex: ROR 30.

3) Signals in maximum mode

1) S_0, S_1, S_2 → status line.

2) $\overline{CS}, \overline{A16}$ → Aves.

2) Pipelined architecture of 8086

→ 8086 supports pipeline architecture.

→ It is a 16 bit microprocessor.

→ 16 bit data bus.

→ has a flag.

→ It has memory bank. Since it has 16 bit data bus.

→ It is faster since it can complete 6 bytes in one cycle.

→ In the pipeline architecture of 8086, it has 6 bytes.

BIV allows two numbers.

- 80386 Architecture Classified into Bus Interface Unit (BIU) and Execution Unit (EU).
- BIU consist of Segment registers such as CS, DS, SS, ES.
- EU consists of General Purpose Register and Pointer Register.

5) 1) Data transfer instructions.

1) `MOV Dest, src` → Moves from destination to same code.

eg: `MOV AX, BX`. (The value of BX is moved to AX)

2) `XCHG Src` → exchange the values.

eg: `XCHG AX`.

3) `PUSH Src` → push value to stack.

eg: `PUSH AX`.

4) `POP Src` → pop values from stack to destination.

eg: `POP AX`.

2) Arithmetic instructions

1) `ADD Dest, src` → Addition of two numbers and store result in dest.

eg: `ADD AX, BX`.

MUL SRC $\rightarrow$ multiplication of a number

eg: MUL AX.

3) DIV SRC $\rightarrow$ Division of a number

eg: DIV AX.

4) IN AL/AX port $\rightarrow$ moves value to port.

eg: ~~AX~~ IN 80H.

5) OUT port AL/AX $\rightarrow$ moves value from port to destination.

eg: 80H OUT.

2) In 8085 Architecture it has following parts.

- 1) Interrupt I/O control $\rightarrow$ control interrupt using signals ^{INT, ~~INT~~} ^{PIST} ^{SSR}
- 2) Serial I/O control $\rightarrow$ control serial communication by signals ^{SIO,} ^{SID.}
- 3) Accumulator $\rightarrow$ performs arithmetic operations and store result ^{Helps}
- 4) Temporary register $\rightarrow$ store a value during arithmetic operations
- 5) Instruction register $\rightarrow$ get instruction.
- 6) Multiplexer array
- 7) Register array
- 8) Stack pointer $\rightarrow$ points to stack.
- 9) Program counter $\rightarrow$ execute next instruction.
- 10) Time and control unit etc....
- 11) ALU $\rightarrow$ performs Arithmetic operations.

ii) ~~MOV~~ ^{CLE} MOV [AX], [BX].
 $\Rightarrow$ MOV 1000, 2000

ii) MOV AX, 5000 [BX]
 $\rightarrow$ ^{CLE} MOV 1000, 5000 2000